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Uniform, large-area self-limiting layer synthesis of tungsten diselenide

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Abstract

A process for the self-limited layer synthesis (SLS) of WSe₂ on SiO₂ substrates has been developed that provides systematic layer number controllability with micrometer-scale (>90%) and wafer-scale (~8 cm) uniformity suitable electronic and optoelectronic device applications. This was confirmed by the fabrication and testing of a WSe₂ back-gated field effect transistor (FET) using Pd (30 nm) as the contact metal, which exhibited p-type behavior with an on/off ratio of ~10⁶ and a field-effect hole mobility of 2.2 cm² V⁻¹ s⁻¹ value, which was higher than has been reported for WSe₂-based FETs produced by conventional chemical vapor deposition. On the basis of these results, it is proposed that the SLS method is universally applicable to a range of device applications.

Introduction

Two-dimensional (2D) transition metal dichalcogenides (TMDCs) have attracted considerable attention [1–7] as a result of their remarkable electronic and optoelectronic properties, which include a unique indirect-to-direct bandgap transition with reducing layer numbers [8], high carrier mobility [9] and strong spin-orbital coupling due to broken inversion symmetry [10]. These exotic properties make 2D TMDCs particularly well suited for use in electronic and optoelectronic devices like photo/gas sensors and field effect transistors (FET) [11]. Indeed, WSe₂ is of particular interest as a p-type substitute for n-type MoS₂, as p-FETs based on exfoliated single-crystal WSe₂ have been demonstrated to have a high hole mobility (~few hundreds of cm² V⁻¹ s⁻¹) and on/off ratio (~10⁸) [12]. Their practical application, however, has so far been limited by a poor wafer-scale uniformity, crystallinity, and layer number controllability. Several studies have succeeded in preparing atomically thin WSe₂ by exfoliation [13], the selenization of metal oxide thin films [14], or chemical vapor deposition (CVD) [15, 16], but these still failed to achieve the necessary large area uniformity

and layer controllability. Consequently, there is still a clear need for an improved method of synthesizing atomically thin WSe₂ for practical device application.

The authors of this paper have previously reported the self-limiting layer synthesis (SLS) of 2D MoS₂ by means of a process in which the layer number is determined by temperature rather than the number of process cycles [17]. This is made possible by the chemically inactive nature of 2D MoS₂, and provides for excellent layer controllability (from mono- to tri-layers) and large area uniformity (over 90%). The present study therefore looks at the feasibility of applying this method to the synthesis of WSe₂ using WCl₆ and diethyl selenide (DESe). By using this liquid-phase organic precursor in place of the gaseous H₂S reactant used with MoS₂, the self-limiting growth characteristics of WSe₂ become purely dependent on temperature after a certain number of process cycles. The layer uniformity, reproducibility, and controllability made by possible by this are herein discussed along with the electrical characteristics of the synthesized WSe₂ when used in a back-gate FET.

Methods

SLS WSe₂ growth

A tube furnace reactor was used to synthesize WSe₂ directly onto SiO₂ (285 nm)/Si substrates using WCl₆ and diethyl selenide (DESe) as the precursor and reactant, respectively. A bubbler containing the precursor was heated to 85 °C to ensure an adequate vapor pressure for the precursor molecules to be carried into the tube by a pure Ar (99.999%) carrier gas. In addition, H₂ gas is injected along with DESe during the reactant time for assisting reduction of WCl₆ and removal of Cl as form of HCl. The SLS cycle consisted of four steps, each with the same with atomic layer deposition (ALD) procedure of: precursor exposure for 4 s, a 5 s Ar purge, 3 s DESe reactant exposure, and a final 5 s Ar purge.

Transfer of WSe₂

The WSe₂ synthesized on the SiO₂ substrates was spin coated with polymethyl methacrylate (PMMA) at 4000 rpm for 60 s. After curing the PMMA at 100 °C for 15 min, the sample was immersed in a 10% HF solution to etch away the SiO₂ layer. The sample was then washed with DI water and transferred to a new SiO₂/Si substrate. Finally, the PMMA was removed using acetone, and the sample was washed with isopropyl alcohol.

Back-Gated field-effect transistor fabrication

A WSe₂-based FET was fabricated from three layers of WSe₂ synthesized on a SiO₂ (300 nm)/Si substrate through the evaporation of Pd (30 nm) electrodes by conventional photolithography and reactive ion (O₂ plasma) etching.

Characterization of WSe₂

The uniformity achieved by the SLS process was assessed by synthesizing three layers of WSe₂ onto 2 × 8 cm² SiO₂ substrates; the substrate size being limited in this instance by the diameter (~3 cm) and length (10 cm hot zone) of the tube furnace used. To assess the cm-scale uniformity, Raman spectra (HORIBA, Lab Ram ARAMIS; 532 nm laser excitation wavelength) were obtained from eight different positions along the length of the sample. The crystallinity and atomic arrangement of the WSe₂ were also evaluated through high-resolution TEM (HRTEM) analysis (FEI Titan G2 Cube 60-300, accelerating voltage = 80 kV). Optical microscopy (Nikon ECLIPSE LV100ND), x-ray photoelectron spectroscopy (XPS: Thermo U.K., K-alpha radiation) and a voltage/current meter (Keithley 4200, Keithley Instruments) were also used in characterizing the WSe₂ produced by SLS.

Results and discussion

Figures 1(a)–(c) presents the AFM images and height profiles obtained from WSe₂ that was synthesized by 100 ALD cycles at growth temperatures of 600 °C, 700 °C and 800 °C. Transferring this WSe₂ to new SiO₂ substrates found that a thickness of 3.9 nm was produced at 600 °C, 2.5 nm at 700 °C and 0.8 nm at 800 °C; these values corresponding to a WSe₂ layer thickness of 5, 3 and 1, respectively. The Raman spectra ($\lambda_{\text{exc}} = 532$ nm) in figure 1(d) show that this 5–1 layer thick WSe₂ exhibits E_{2g}^1 and A_{1g} modes from in-plane and out-of-plane vibrations at around 251 and 258 cm⁻¹. The characteristic A_{1g}^2 (out-of-plane mode) peak that is observed at ~308 cm⁻¹ with the 3 and 5 layer WSe₂ has previously only been observed in WSe₂ when it is a bilayer or thicker, thereby confirming that the WSe₂ synthesized at 800 °C is a monolayer. As the change in the relative peak intensity ratio (E_{2g}^1/A_{1g}) is also dependent on the layer number, the Raman peaks were fitted using a Lorentzian function to extract the peak intensities for the E_{2g}^1 and A_{1g} modes. Figure 1(e) shows the relative peak intensity ratio as a function of the layer number, in which we see that the intensity ratio decreases from 6.56 with a monolayer to 2.31 with five layers of WSe₂. This can be attributed to a change in scattering volume, which has been seen with other TMDCs [18–20].

The optical characteristics of the synthesized WSe₂ were evaluated as a function of layer number through PL measurement (figure 2(a)), which revealed that a monolayer exhibits a strong PL signal at 1.63 eV that corresponds to direct excitonic transitions of WSe₂ [10, 13]. This signal weakens and becomes negligible as the number of layers is increased, as this induces a transition from a direct to an indirect band gap. This is concordant with previous results regarding the dependence of the PL signal on the number of layers, and further confirms the temperature dependent nature of growth by SLS. In the XPS spectra of the monolayer WSe₂ synthesized at 600 °C shown in figure 2(b), we see that the W4f core level has peaks at 38.4, 35.1 and 32.9 eV that correspond to the W5p_{3/2}, W4f_{5/2} and W4f_{7/2} levels, respectively. The Se3d core level exhibits two peaks at 55.9 and 55.1 eV, which correspond to the Se3d_{5/2} doublet and Se3d_{3/2}, respectively [15, 21, 22]. The Cl2p core level exhibits no discernible peaks, indicating that the WSe₂ contained no XPS-detectable Cl species. From this, the stoichiometric ratio of Se:W was calculated to be 2.

The effect that the number of process cycles has on the number of WSe₂ layers produced by SLS was investigated using 100, 150 and 200 cycles at growth temperatures of 600 °C, 700 °C and 800 °C. The resulting thickness, as measured by AFM (figure S1), is plotted as a function of the number of cycles in figure 3(a). This clearly shows that the number of WSe₂ layers does not increase linearly with the number of cycles,

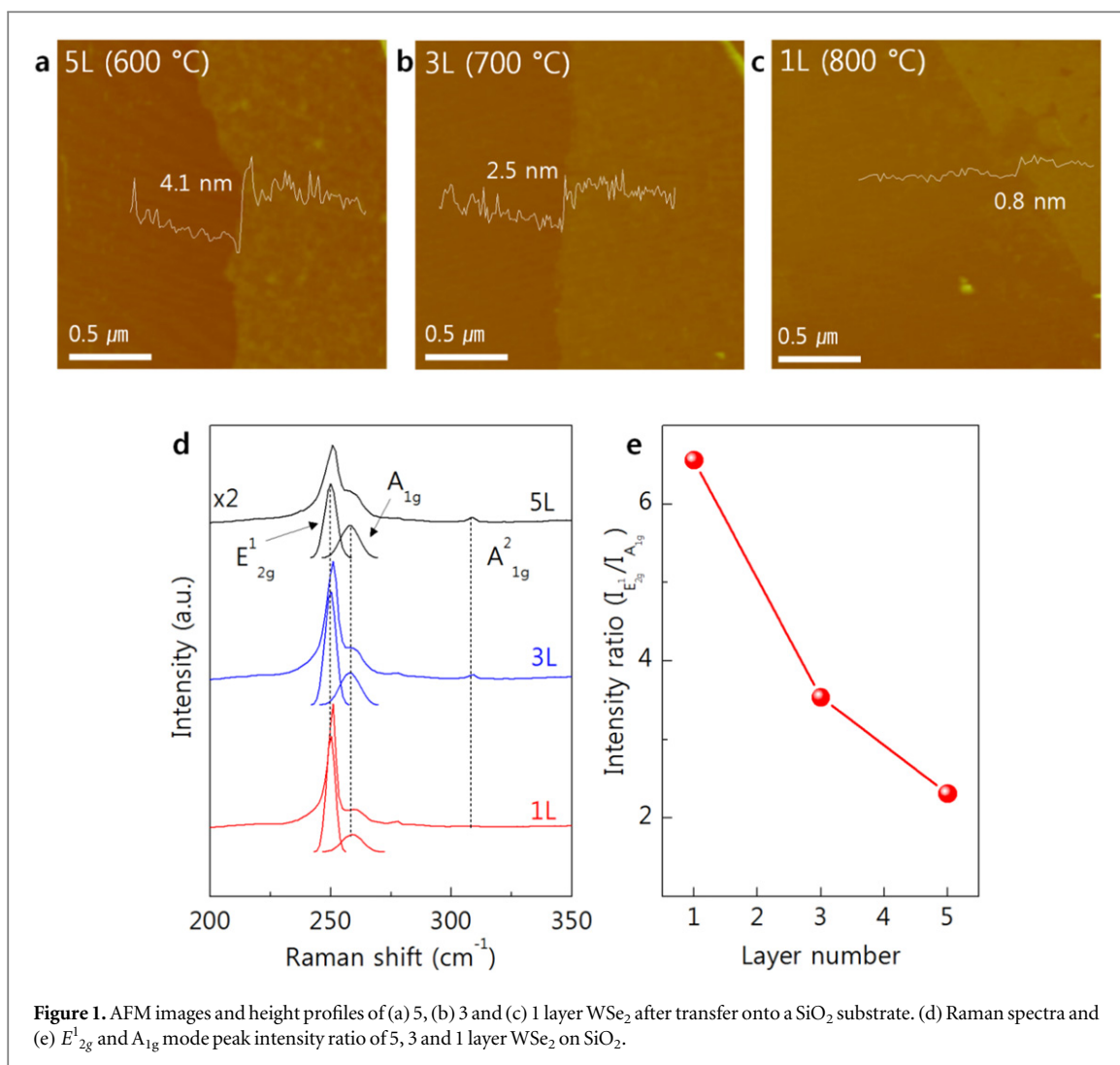


Figure 1. AFM images and height profiles of (a) 5, (b) 3 and (c) 1 layer WSe₂ after transfer onto a SiO₂ substrate. (d) Raman spectra and (e) E_{2g}^1 and A_{1g} mode peak intensity ratio of 5, 3 and 1 layer WSe₂ on SiO₂.

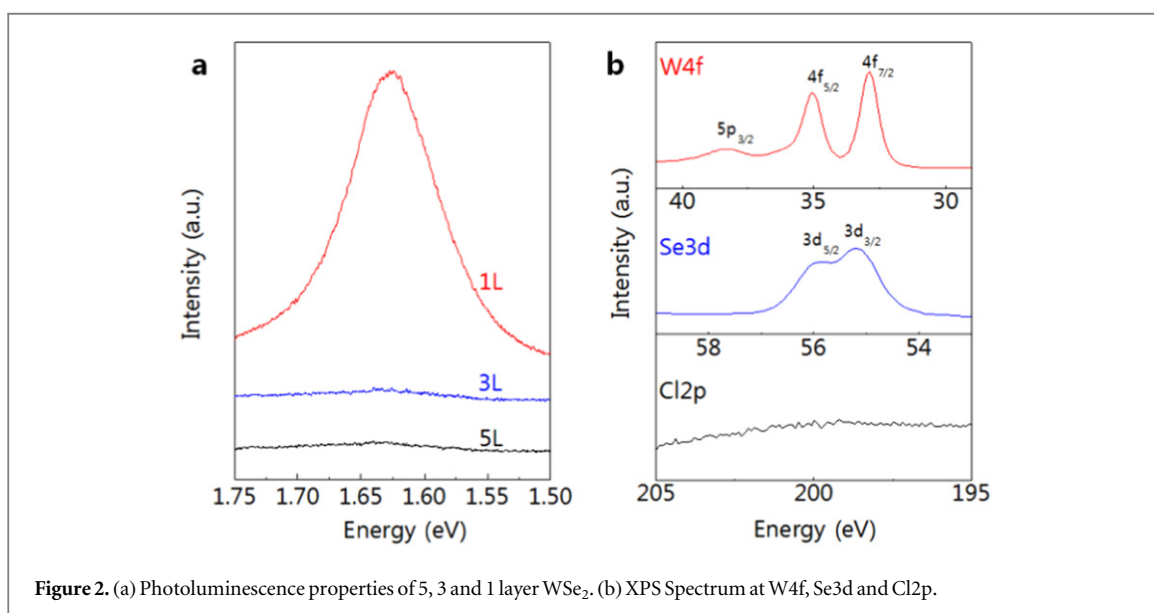


Figure 2. (a) Photoluminescence properties of 5, 3 and 1 layer WSe₂. (b) XPS Spectrum at W4f, Se3d and Cl2p.

but rather saturates at a certain critical point determined by the growth temperature (800 °C, 700 °C and 600 °C for 1, 3 and 5 layers, respectively). This stands in stark contrast to the usual growth characteristics of

the ALD process, where thickness does increase linearly with increasing process cycles, and so clearly there is a different growth mechanism in effect during the SLS process.

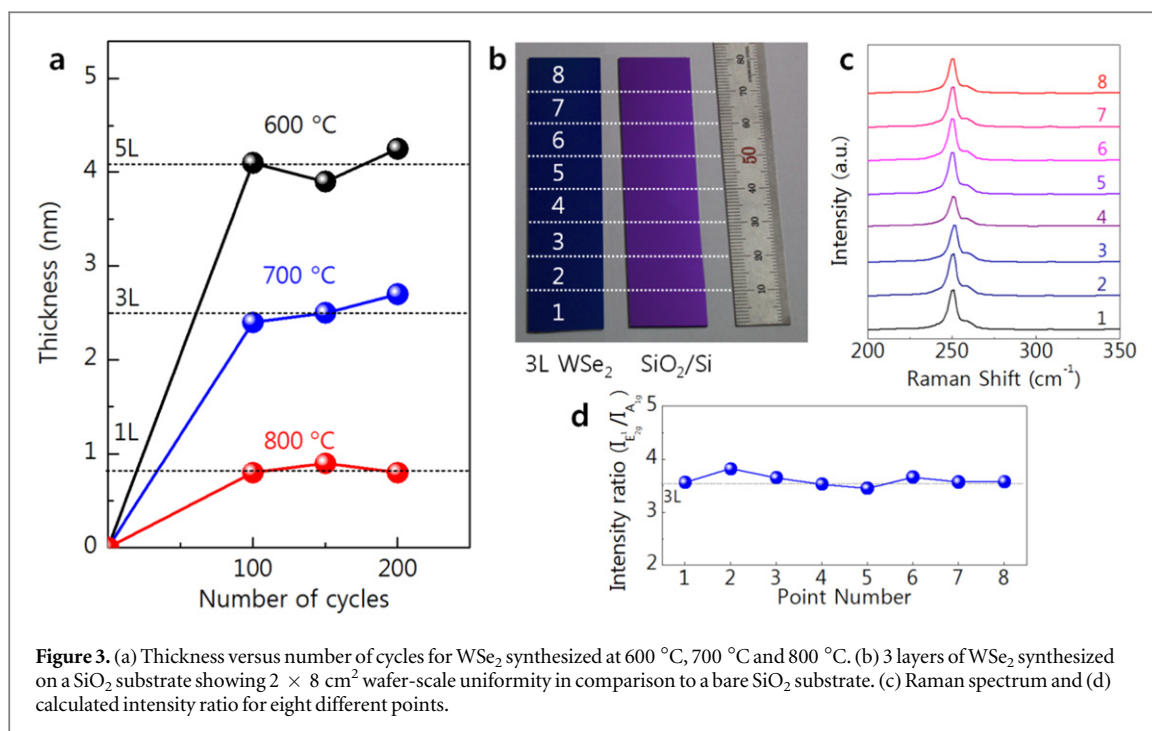


Figure 3. (a) Thickness versus number of cycles for WSe₂ synthesized at 600 °C, 700 °C and 800 °C. (b) 3 layers of WSe₂ synthesized on a SiO₂ substrate showing 2 × 8 cm² wafer-scale uniformity in comparison to a bare SiO₂ substrate. (c) Raman spectrum and (d) calculated intensity ratio for eight different points.

This self-limiting aspect of the SLS process is believed to be the result of the same mechanisms previously identified in the SLS of MoS₂, which can be summarized as follows. Firstly, the fact that the surface of a TMDC is inherently chemically inert forces synthesis to proceed through physical adsorption [23, 24]. Secondly, the surface potential depth of WCl₆ precursor molecules decreases with an increasing number of WSe₂ layers due to their screening effect on the electric field [25]. Consequently, once a certain number of WSe₂ layers have been formed at any given growth temperature with sufficiently high thermal energy, most of the WCl₆ molecules that are physically adsorbed on the WSe₂ basal plane can be easily desorbed due to the reduced potential depth [23, 24]. This creates the same self-limiting growth behavior seen in the synthesis of MoS₂ using a gaseous reactant, and shows that the process can be just as easily used with a liquid-phase organic reactant. It therefore stands to reason that this SLS process could be applied to the synthesis of a wide range of two-dimensional TMDCs, regardless of the chalcogen reactant needed. It is noteworthy that our SLS WSe₂ mechanism is totally different from the previous reported self-limiting CVD MoS₂, which is limited by loaded dose of the precursor.

The wafer-scale (2 × 8 cm²) uniformity achieved by SLS is demonstrated by the consistency in color of the WSe₂ shown in figure 3(b), with the Raman spectra in figure 3(c) also showing very little variation between the eight different points tested. The E_{2g}^1 and A_{1g} peaks in figure 3(c) were fitted to obtain the ratio between the two for each point, which are plotted in figure 3(d). This found that the intensity ratio was consistently low, with an average value of 3.6 and a standard deviation of 0.11.

The low-magnification TEM image of the tri-layered WSe₂ in figure 4(a) reveals that there are some pale-contrast regions with fewer layers; but as these represent only about 2% of the total area, the micrometer-scale layer uniformity can be considered near-perfect (i.e., >90%). The approximate grain size is 200 nm as presented using red dashed triangle. The HRTEM image in figure 4(b) shows Moiré pattern that result from 3L WSe₂ growth. The diffraction pattern in the fast Fourier transform (FFT) image in figure 4(c) clearly shows the six-fold coordination symmetry of the three different layers with angles of 27° and 12°. The inverse FFT (IFFT) of the pattern in red that is given in figure 4(d) reveals a honeycomb-like structure, with lattice spacings of 0.28 and 0.17 nm for the (100) and (110) planes, respectively.

The room temperature electrical performance of the back-gated SLS WSe₂ FET under atmospheric pressure was calculated from the linear regime of the transfer curve shown in figure 5(a) using the equation:

$$\mu = \left(\Delta I_d / \Delta V_g \right) \times L / \left(W C_{ox} V_d \right),$$

where L , W , and C_{ox} are the channel length, channel width and gate capacitance per unit area, respectively. This transfer curve is consistent with p-type behavior, with the field effect hole mobility ($2.2 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) and on/off current ratio (10^6) extracted from this being significantly higher than previously reported values for a WSe₂ back-gated FET device produced using CVD as the method of synthesis [26].

Conclusion

This study has succeeded in achieving self-limiting layer synthesis of WSe₂ on a SiO₂ substrate, using a

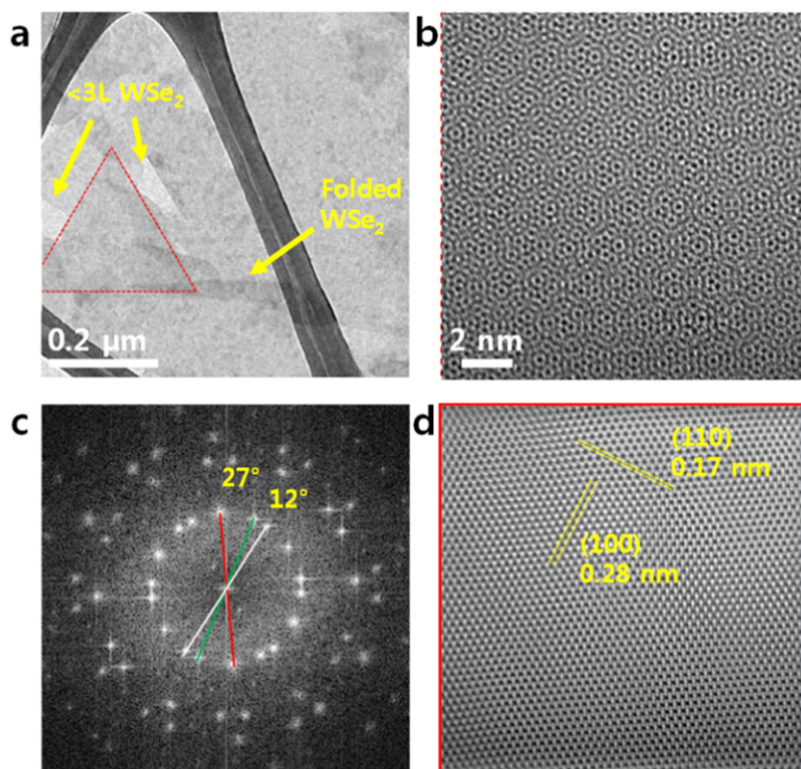


Figure 4. (a) Low resolution TEM image of three layers of WSe₂ on a TEM grid with red dashed triangle on single grain and (b) HRTEM image of selected region. (c) Corresponding FFT pattern and (d) honeycomb structure of a single layer extracted by IFFT from the FFT pattern.

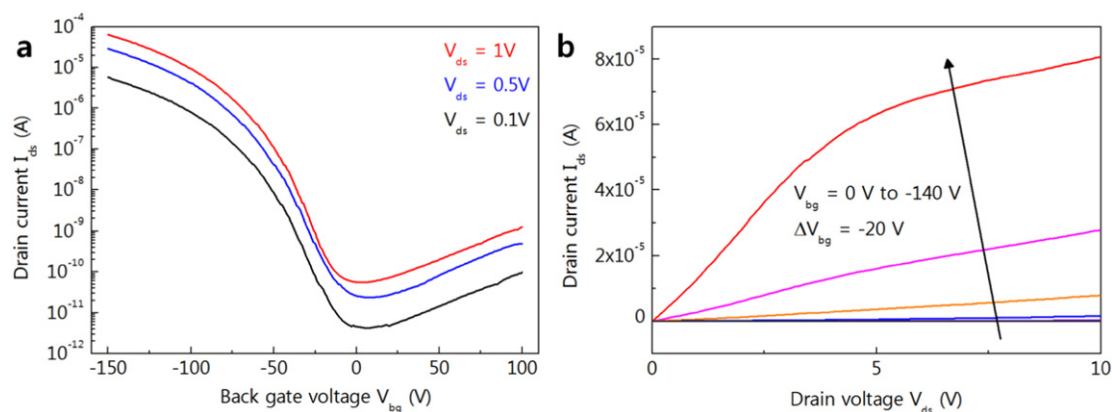


Figure 5. (a) Transfer curve and (b) output curve of a FET fabricated from three-layer thick WSe₂.

method in which the number of layers is controlled solely by varying the growth temperature to produce a high layer uniformity at both the micrometer-scale (>90%) and wafer-scale (~8 cm). When incorporated into a FET using Pd (30 nm) as the ohmic contact metal [27], this SLS WSe₂ produced excellent p-type behavior with a hole mobility of $2.2 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and on/off ratio of $\sim 10^6$. This SLS method therefore has great potential to be universally applied to the fabrication of a range of practical devices.

Acknowledgments

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